

GUIDELINE DOCUMENT — FOR ADOPTION

BIM GUIDELINE FOR
HALLS OF WORSHIP

DOCUMENT NO. GMF-BIM-001	VERSION 1.0 — Draft for adoption	DATE 04 July 2026
APPLIES TO All design and as-built models of Halls of Worship and ancillary buildings	STATUS Mandatory once adopted	PREPARED BY Vollmann Akarakiri

01. PURPOSE

Every Hall of Worship must have a **single, complete BIM model** that serves as the source of truth for its design, construction and future maintenance. The goals:

- 01

Drawings come from the model — plans, sections, elevations, schedules and quantities are extracted, never drawn separately, so they can never disagree with each other.
- 02

Accurate quantities — the Bill of Quantities is taken off the model, which removes a major source of cost overruns.

- 03

Reusability — a modelled hall becomes a template library for the next project instead of starting from zero.
- 04

A permanent record — as-built models support maintenance, extensions and repairs for decades.

02. SOFTWARE AND FILE FORMATS

Item	Standard
Authoring tool	Any capable BIM authoring tool (e.g. Revit, ArchiCAD, FreeCAD/BlenderBIM). One tool per project , agreed at Gate 1 and recorded in the BIM Execution Plan.
Exchange / archive format	IFC 2×3 or IFC 4 — every issued model must also be exported to IFC so it remains readable regardless of software.
Drawings issue format	PDF (signed issues) + DWG/DXF where a contractor requests it.
Coordinates & units	Real-world site coordinates from the licensed survey plan; project base point at a marked, recoverable site benchmark. Units: millimetres ; elevations in metres above datum.

Native files, IFC exports and PDFs are all archived — never only the native file.

03. MODEL STRUCTURE

- **One federated project, discipline models separated:** Architecture, Structure, MEP (mechanical / electrical / plumbing), Site / External works.
- Each discipline model is linked, not merged, and clash-checked before every gate (Section 06).
- Storeys/levels named identically in all discipline models: **L00 Ground, L01 First, RF Roof, FD Foundation**.

04. NAMING CONVENTIONS

4.1 — PROJECT CODE

As defined in GMF-PG-001 §6: **GMF-<STATE>-<TOWN>-<NN>**, e.g. **GMF-LA-LAGOS-01**.

4.2 — FILE NAMES

<ProjectCode>-<Originator>-<Discipline>-<Type>-<Number>-<Revision>

Field	Codes
Originator	Short code for the firm/person, e.g. ADO
Discipline	AR architecture · ST structure · ME mechanical · EL electrical · PL plumbing · SI site · ZZ federated
Type	M3 model · DR drawing · SC schedule · RP report · BQ bill of quantities
Revision	P01 , P02 ... preliminary · C01 , C02 ... issued for construction · A01 ... as-built

Example: **GMF-EN-ENUGU-01-ADO-AR-M3-0001-P03.ifc**

4.3 — DRAWING NUMBERS

<ProjectCode>-<Discipline>-<Series><Sheet>

Series: **0** general/site · **1** plans · **2** elevations · **3** sections · **4** details · **5** schedules · **6** MEP layouts. Example: **GMF-EN-ENUGU-01-AR-101** = architecture, ground floor plan.

05. LEVEL OF DEVELOPMENT (LOD) BY PHASE

Project phase (GMF-PG-001)	Required LOD	Meaning
Phase 1 — Feasibility	LOD 100	Massing, site placement, capacity check
Phase 2 — Concept design	LOD 200	Approximate geometry, key spaces sized
Phase 2 — Technical design (Gate 2)	LOD 300	Precise geometry; quantities can be taken off
Phase 3–4 — Construction	LOD 350	Coordination detail, connections, penetrations
Phase 5 — Handover	LOD 500	Verified as constructed, with equipment data (as-built)

Minimum modelled content at Gate 2 (LOD 300): foundations, all structural members, walls with true build-ups, roof structure and covering, doors/windows with schedules, floor and

ceiling finishes, sanitary fittings, major electrical (distribution boards, generator, lighting layout), water storage and drainage, external works within the site boundary.

06. QUALITY CHECKS BEFORE EVERY ISSUE

1. **Clash detection** between all discipline models — zero unresolved hard clashes (structure vs. services vs. architecture).
2. **Naming audit** — files, levels and drawing numbers per Section 04.
3. **Quantity sanity check** — QS compares model take-off against manual spot checks on at least concrete, blockwork and roofing.
4. **Model checklist** signed by the Design Lead and filed with the gate submission.

07. STANDARD FOLDER STRUCTURE

Every project repository/drive uses exactly this structure:

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GMF-<STATE>-<TOWN>-<NN>/
├─ 01-Admin/           (charter, minutes, approvals, contracts)
├─ 02-Brief/           (capacity brief, feasibility, surveys, soil report)
├─ 03-Models/
│   ├─ native/         (authoring files)
│   └─ ifc/            (issued IFC exports, one per gate)
├─ 04-Drawings/
│   ├─ issued/         (signed PDF issues, by revision)
│   └─ superseded/
├─ 05-Costs/           (cost plans, BOQ, valuations, certificates – GMF-CM-001)
├─ 06-Site/            (site diaries, weekly reports, photos, instructions)
├─ 07-Handover/        (as-built model, O&M info, warranties, final account)
└─ 08-Lessons/         (lessons-learned report)
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08. THE HALL TEMPLATE LIBRARY

- After each project closes, reusable components (seating layouts, door/window families, roof truss types, altar/platform arrangements, standard finishes) are copied into a shared **template library** maintained alongside this guideline.
- New projects **must start from the template library**, adapting to site and capacity, rather than modelling from scratch.
- A standard capacity series should be developed over time (e.g. 150-seat, 300-seat, 500-seat reference halls) so that a new project begins at LOD 200 on day one.

09. BIM EXECUTION PLAN (BEP)

At Gate 1 the Design Lead submits a one-to-two page BEP recording: authoring software and versions, model originators per discipline, exchange schedule, coordinate/benchmark definition, and who runs clash detection. The BEP is approved by the PM and filed in 01-Admin/.

10. AS-BUILT REQUIREMENT

Final payment to the design team and contractor is conditional (GMF-PG-001 §6) on delivery of:

- as-built model (native + IFC) updated to what was actually constructed,
- as-built drawings (PDF),
- equipment schedules with make/model/capacity for generator, pumps, tanks, distribution boards and fixed AV equipment.

11. REVISION HISTORY

Version	Date	Change	Author
1.0	04 July 2026	First issue for adoption	Vollmann Akarakiri

AUTHORISATION

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